AGRO SENSOR READINGS

Stores sensor data collected from agricultural fields at regular time intervals
Each document represents one sensor reading from one device at a specific date and time

Contains environmental data such as:
- Temperature
- Humidity
- Rainfall
- light intensity

Includes soil-related parameters:
- soil moisture
- soil pH
- fertilizer usage

Tracks crop-related information:
- crop type
- season
- crop health status
- estimated yield
- pest risk level

Helps identify irrigation requirements using the *irrigation_needed* field
Uses *anomaly_flag* to mark abnormal or unusual sensor readings
device_id and *location* fields allow comparison across different regions and sensors.

Separate date and time fields support daily analysis and time-based queries.

agro_sensor_readings

_id	objectId
anomaly_flag	bool
crop_health	string
crop_type	string
date	date
device_id	string
fertilizer_us…	double
humidity	double
irrigation_ne…	bool
light_intensi…	int
location	string
pest_risk	string
rainfall	double
season	string
soil_moisture	double
soil_ph	double
temperature	double
time	string
timestamp	string
yield_estimate	double

Overall document sample:


```
_id: ObjectId('697646e381e5b12473c11840')
timestamp : "14/03/24 19:00"
device_id : "BD-KHU-005"
location : "Khulna"
crop_type : "Wheat"
season : "summer"
temperature : 30.3
humidity : 75.29
rainfall : 4.39
soil_moisture : 49.27
soil_ph : 7.29
light_intensity : 17
fertilizer_used : 29.39
irrigation_needed : false
crop_health : "Healthy"
yield_estimate : 3.5
pest_risk : "Medium"
anomaly_flag : false
date : 2024-03-14T00:00:00.000+00:00
time : "19:00"
```

QUERY PERFORMED FOR PERFORMANCE MEASUREMENT.

```
db.agro_sensor_readings.find({
  location: "Dhaka",
  date: {
    $gte: ISODate("2024-03-01T00:00:00Z"),
    $lt: ISODate("2024-04-01T00:00:00Z")
  }
}).explain("executionStats")
```

```
    nReturned: 265,
    executionTimeMillis: 7,
```

```
db.agro_sensor_readings.createIndex({ location: 1, date: 1 })
```

```
  nReturned: 265,
  executionTimeMillis: 3,
```

Other indexing created:

```
db.agro_sensor_readings.createIndex({ date: 1 })
db.agro_sensor_readings.createIndex({ device_id: 1, date: -1 })
db.agro_sensor_readings.createIndex({ location: 1, date: 1 })
db.agro_sensor_readings.createIndex({ crop_type: 1, date: 1 })
db.agro_sensor_readings.createIndex({ anomaly_flag: 1, date: 1 })
```

PARAM DHOLAKIA

Name & Definition	Type	Size	Usage	Properties	Status
_id_	REGULAR	114.7 kB	7 (since Sun Jan 25 2026)	UNIQUE	READY
location_1_date_1	REGULAR	135.2 kB	0 (since Mon Jan 26 2026)	COMPOUND	READY
date_1	REGULAR	131.1 kB	0 (since Mon Jan 26 2026)		READY
device_id_1_date_-1	REGULAR	139.3 kB	0 (since Mon Jan 26 2026)	COMPOUND	READY
crop_type_1_date_1	REGULAR	135.2 kB	0 (since Mon Jan 26 2026)	COMPOUND	READY
anomaly_flag_1_date_1	REGULAR	131.1 kB	0 (since Mon Jan 26 2026)	COMPOUND	READY

1 Daily average temperature per location

```
db.agro_sensor_readings.aggregate([
  {
    $group: {
      _id: { location: "$location", date: "$date" },
      avgTemp: { $avg: "$temperature" }
    }
  },
  { $sort: { "_id.date": 1 } }
])
```

What it does:
Shows how average temperature changes daily for each city.
Useful for climate trend charts.

```
  _id: {
    location: 'Sylhet',
    date: 2024-01-01T00:00:00.000Z
  },
  avgTemp: 21.853
}
{
  _id: {
    location: 'Khulna',
    date: 2024-01-01T00:00:00.000Z
  },
  avgTemp: 20.853333333333335
}
```

---

2 Total rainfall per location (monthly)

```
db.agro_sensor_readings.aggregate([
 {
  $group: {
   _id: { location: "$location" },
   totalRainfall: { $sum: "$rainfall" }
  }
 }
])
```

What it does:
Calculates how much rain each location received.
Helps compare water availability across regions.

```
    _id: {
      location: 'Chattogram'
    },
    totalRainfall: 5551.85
  }
  {
    _id: {
      location: 'Rajshahi'
    },
    totalRainfall: 5415.67
  }
```

---

3 Latest sensor reading per device

```
db.agro_sensor_readings.aggregate([
 { $sort: { date: -1 } },
 {
  $group: {
    _id: "$device_id",
    latestReading: { $first: "$$ROOT" }
  }
 },
 {
  $project: {
    _id: 0,
    device_id: "$_id",
    date: "$latestReading.date",
    location: "$latestReading.location",
    temperature: "$latestReading.temperature",
    soil_moisture: "$latestReading.soil_moisture",
    crop_health: "$latestReading.crop_health"
  }
 }
```

PARAM DHOLAKIA

])**What it does:**

Fetches the most recent data from every sensor device.
Used for live monitoring dashboards.

```
{
  device_id: 'BD-CHA-008',
  date: 2024-06-28T00:00:00.000Z,
  location: 'Chattogram',
  temperature: 30.59,
  soil_moisture: 15.11,
  crop_health: 'Critical'
}
```

---

4 Number of anomalies per location

```
db.agro_sensor_readings.aggregate([
 { $match: { anomaly_flag: true } },
 {
  $group: {
   _id: "$location",
   anomalyCount: { $sum: 1 }
  }
 }
])
```

What it does:

Counts abnormal sensor readings by city.
Helps identify problematic regions quickly.

```
< {
    _id: 'Khulna',
    anomalyCount: 30
  }
  {
    _id: 'Chattogram',
    anomalyCount: 38
  }
```

5 Crops currently under stress

```
db.agro_sensor_readings.find(
  { crop_health: "Stressed" },
  { crop_type: 1, location: 1, date: 1 }
)
```

What it does:
Lists crops that are not healthy.
Useful for alerting farmers or agronomists.

```
{
  _id: ObjectId('697646e381e5b12473c10952'),
  location: 'Rajshahi',
  crop_type: 'Jute',
  date: 2024-01-06T00:00:00.000Z
}
{
  _id: ObjectId('697646e381e5b12473c10956'),
  location: 'Dhaka',
  crop_type: 'Wheat',
  date: 2024-01-06T00:00:00.000Z
}
```

6 Irrigation required today

```
db.agro_sensor_readings.find(
  {
    irrigation_needed: true,
    date: {
      $gte: ISODate("2024-03-15T00:00:00Z"),
      $lt: ISODate("2024-03-16T00:00:00Z")
    }
        },
```

```
  {
    "_id":0, "crop_type": 1, "location": 1, "device_id":1
  }
)
```

What it does:
Shows which farms need watering today.
Helps optimize irrigation planning.

```
< {
    device_id: 'BD-CHA-001',
    location: 'Chattogram',
    crop_type: 'Jute'
  }
  {
    device_id: 'BD-DHA-002',
    location: 'Dhaka',
    crop_type: 'Jute'
  }
```

---

7 Average yield estimate per crop

```
db.agro_sensor_readings.aggregate([
  {
    $group: {
      _id: "$crop_type",
      avgYield: { $avg: "$yield_estimate" }
    }
  }
])
```

What it does:
Calculates expected yield for each crop type.
Useful for production forecasting.

```
< {
    _id: 'Jute',
    avgYield: 2.6615773508594542
  }
  {
    _id: 'Potato',
    avgYield: 23.872422577422576
  }
```

---

8 Pest risk distribution

```
db.agro_sensor_readings.aggregate([
 {
  $group: {
   _id: "$pest_risk",
   count: { $sum: 1 }
  }
 }
])
```

What it does:
Shows how many readings fall under low/medium/high pest risk.
Useful for preventive pest control decisions.

```
< {
    _id: 'Medium',
    count: 3267
  }
  {
    _id: 'Low',
    count: 6733
  }
```

---

9 Devices with frequent anomalies

```
db.agro_sensor_readings.aggregate([
 { $match: { anomaly_flag: true } },
```

```
  {
    $group: {
      _id: "$device_id",
      anomalyCount: { $sum: 1 }
    }
  },
  { $sort: { anomalyCount: -1 } }
])
```

What it does:
Identifies faulty or problematic sensor devices.
Helps in maintenance and sensor calibration.

```
{
  _id: 'BD-KHU-002',
  anomalyCount: 5
}
{
  _id: 'BD-RAJ-002',
  anomalyCount: 5
}
```

---

10 Soil moisture trend over time

```
db.agro_sensor_readings.aggregate([
  {
    $group: {
      _id: "$date",
      avgMoisture: { $avg: "$soil_moisture" }
    }
  },
  { $sort: { "_id": 1 } }
])
```

What it does:
Tracks soil moisture changes day by day.
Useful for irrigation and drought analysis.

```
{
  _id: 2024-01-19T00:00:00.000Z,
  avgMoisture: 55.212857142857146
}
{
  _id: 2024-01-20T00:00:00.000Z,
  avgMoisture: 54.79035714285715
}
```

---

11 Crop performance by season

```
db.agro_sensor_readings.aggregate([
 {
  $group: {
   _id: { crop: "$crop_type", season: "$season" },
   avgYield: { $avg: "$yield_estimate" }
  }
 }
])
```

What it does:
Compares how crops perform across seasons.
Helps decide optimal planting time.

```
{
  _id: {
    crop: 'Rice',
    season: 'winter'
  },
  avgYield: 4.517837037037037
}
```

---

12 Heat stress detection

```
db.agro_sensor_readings.find({
 temperature: { $gt: 35 }
```

```
}
{
  "_id":0 , "location":1
}
)
```

What it does:
Finds periods of extreme heat.
Useful for heat-stress alerts and crop protection.

```
{
  location: 'Dhaka'
}
{
  location: 'Chattogram'
}
```

---

13 Light intensity vs crop health

```
db.agro_sensor_readings.aggregate([
 {
  $group: {
   _id: "$crop_health",
   avgLight: { $avg: "$light_intensity" }
  }
 }
])
```

What it does:
Analyzes relationship between sunlight and crop health.
Helps in greenhouse or shading decisions.

```
< {
    _id: 'Healthy',
    avgLight: 11964.47452143435
  }
  {
    _id: 'Stressed',
    avgLight: 12808.254178957719
  }
  {
    _id: 'Critical',
    avgLight: 12924.908759124088
  }
```

14 Location-wise average soil pH

```
db.agro_sensor_readings.aggregate([
  {
    $group: {
      _id: "$location",
      avgSoilPH: { $avg: "$soil_ph" }
    }
  }
])
```

What it does:
Shows soil acidity by location.
Useful for fertilizer and soil treatment planning.

```
< {
    _id: 'Rajshahi',
    avgSoilPH: 6.3826318909306465
  }
  {
    _id: 'Chattogram',
    avgSoilPH: 6.389747058823529
  }
```

---

15 Yield trend over time

```
db.agro_sensor_readings.aggregate([
  {
    $group: {
      _id: "$date",
      avgYield: { $avg: "$yield_estimate" }
    }
  },
  { $sort: { "_id": 1 } }
])
```

PARAM DHOLAKIA

What it does:

Tracks how crop yield changes over time.

Useful for long-term agricultural planning.

```
{
  _id: 2024-01-13T00:00:00.000Z,
  avgYield: 11.509107142857143
}
{
  _id: 2024-01-14T00:00:00.000Z,
  avgYield: 13.137818181818183
}
{
  _id: 2024-01-15T00:00:00.000Z,
  avgYield: 17.93267857142857
}
```